

Recurrence Quantification Analysis

Code ▾

Data Exploration

This dataset originates from the traffic signal system, published via the Victoria (Australia) open transport data platform, and contains traffic volumes recorded every 15 minutes throughout March 2026 for multiple intersections. For this group project, we use the combined approach, where detector values are aggregated into a single total traffic volume per site at each timestamp. Detectors are individual sensors at an intersection, typically assigned to specific lanes or traffic movements, that count how many vehicles pass during each 15-minute interval. By summing these detector readings, we obtain a clean and consistent site level time series for analyzing temporal patterns and comparing traffic flow across sites.

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```
library(readr)
library(here)

library(ggplot2)

library(nonlinearTseries)

library(tseriesChaos)

library(crqa)
```

Hide

```
traffic_data <- read_csv(here("data", "traffic_detectors_combined_100_161_164.csv"))
```

Below is a preview of the dataset. Each row represents a 15-minute interval at a specific site, where traffic volumes from all detectors have been aggregated into a single site-level measurement. The key variables are `site_total_volume` , which provides the traffic count and serves as the main time series for analysis, `timestamp` , which ensures the correct temporal order and regular spacing of observations, and `site_no` , which allows comparison of traffic dynamics across different intersections.

Hide

```
head(traffic_data)
```

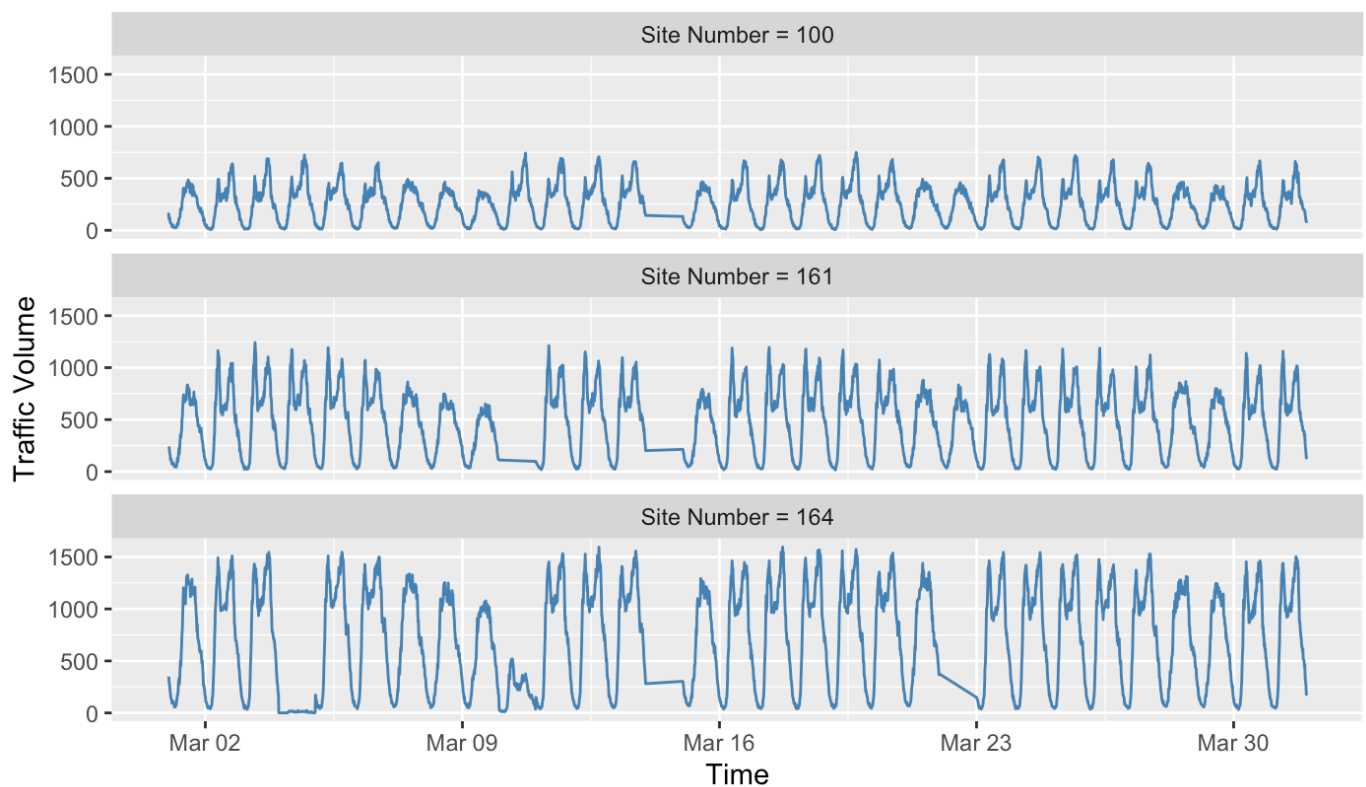
timestamp	date	day_na...	is_weekend	site_no	region	site_total_volum
<S3: POSIXct>	<date>	<chr>	<lgl>	<dbl>	<chr>	<dbl>
2026-03-01 00:00:00	2026-03-01	Sunday	TRUE	100	WV1	16
2026-03-01 00:00:00	2026-03-01	Sunday	TRUE	161	WV1	24
2026-03-01 00:00:00	2026-03-01	Sunday	TRUE	164	WV1	34
2026-03-01 00:15:00	2026-03-01	Sunday	TRUE	100	WV1	11
2026-03-01 00:15:00	2026-03-01	Sunday	TRUE	161	WV1	23
2026-03-01 00:15:00	2026-03-01	Sunday	TRUE	164	WV1	31

6 rows

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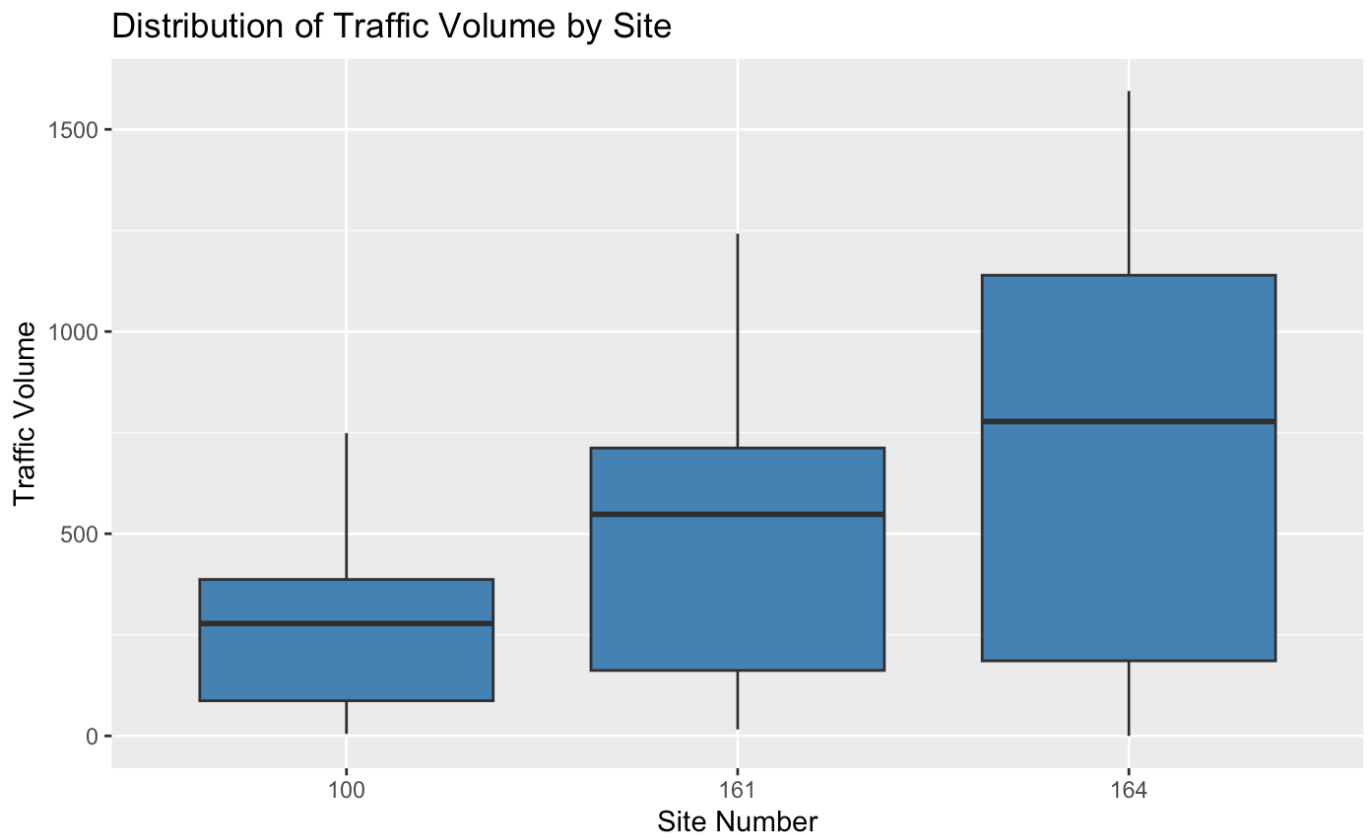
```
ggplot(traffic_data, aes(x = timestamp, y = site_total_volume)) +
  geom_line(color = "steelblue") +
  facet_wrap(~ site_no, ncol = 1, labeller = labeller(site_no = function(x) paste("Site Number =", x)))) +
  labs(
    title = "Traffic Volume Over Time by Site",
    x = "Time",
    y = "Traffic Volume"
  )
```

Traffic Volume Over Time by Site



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```
ggplot(traffic_data, aes(x = factor(site_no), y = site_total_volume)) +
  geom_boxplot(fill = "steelblue") +
  labs(
    title = "Distribution of Traffic Volume by Site",
    x = "Site Number",
    y = "Traffic Volume"
  )
```

[Hide](#)

```
by(traffic_data$site_total_volume, traffic_data$site_no, summary)
```

```
traffic_data$site_no: 100
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   5.0   87.0   278.0   266.3   387.0   749.0
```

```
traffic_data$site_no: 161
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  16.0  162.0   548.0   491.5   712.0  1242.0
```

```
traffic_data$site_no: 164
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.0  185.8   777.5   716.1  1139.2  1595.0
```

RQA

RQA - Single Site Analysis

Based on the data exploration, site 164 was selected for further analysis. This site exhibits the highest variability, with an interquartile range from 185.8 to 1139.2, as well as the highest median traffic volume (Mdn = 777.5), compared to site 161 (Mdn = 548.0) and site 100 (Mdn = 278.0). Furthermore, the observed temporal data exhibit a regularity, which is expected to result in meaningful recurrence patterns.

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```
# Select purely site 164 data
site164 <- traffic_data[traffic_data$site_no == 164, ]

# They should be ordered by just in case
site164 <- site164[order(site164$timestamp), ]

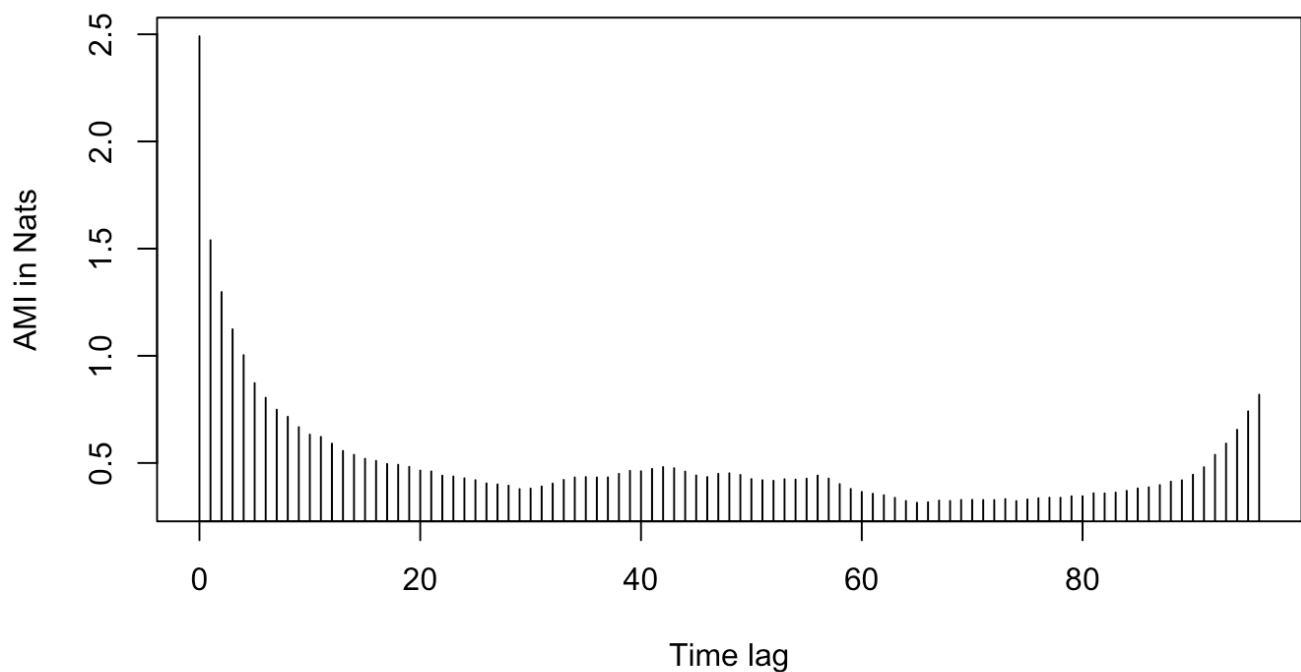
# Extract volumes
x <- site164$site_total_volume
```

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```
x_scaled <- scale(x)[, 1]

ami <- mutualInformation(x_scaled, lag.max = 96, do.plot = TRUE)
```

Average Mutual Information (AMI)

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```
delay <- 29

print(data.frame(lag = 0:96, ami = ami$mutual.information))
```

	lag	ami
1	0	2.4906428
2	1	1.5392889
3	2	1.2975494
4	3	1.1237068
5	4	1.0031715
6	5	0.8726522
7	6	0.8044887
8	7	0.7486949
9	8	0.7156125
10	9	0.6675323
11	10	0.6323387
12	11	0.6218610
13	12	0.5904642
14	13	0.5562657
15	14	0.5381596
16	15	0.5197466
17	16	0.5098468
18	17	0.4953585
19	18	0.4915607
20	19	0.4819943
21	20	0.4648372
22	21	0.4608461
23	22	0.4409571
24	23	0.4373134
25	24	0.4287162
26	25	0.4201276
27	26	0.4050764
28	27	0.4000258
29	28	0.3942372
30	29	0.3783352
31	30	0.3804048
32	31	0.3908023
33	32	0.4040758
34	33	0.4212590
35	34	0.4326887
36	35	0.4345508
37	36	0.4327588
38	37	0.4331413
39	38	0.4491534
40	39	0.4633942
41	40	0.4615130
42	41	0.4724892
43	42	0.4811055
44	43	0.4756030
45	44	0.4599408
46	45	0.4418536
47	46	0.4345817
48	47	0.4497168
49	48	0.4519656
50	49	0.4443022
51	50	0.4252330
52	51	0.4198391
53	52	0.4170452
54	53	0.4242732

55	54	0.4225913
56	55	0.4269457
57	56	0.4415163
58	57	0.4273956
59	58	0.4015509
60	59	0.3789802
61	60	0.3651417
62	61	0.3566737
63	62	0.3497972
64	63	0.3377302
65	64	0.3229709
66	65	0.3148686
67	66	0.3166890
68	67	0.3245832
69	68	0.3229828
70	69	0.3276959
71	70	0.3282822
72	71	0.3272981
73	72	0.3272812
74	73	0.3318636
75	74	0.3224253
76	75	0.3300233
77	76	0.3360037
78	77	0.3381280
79	78	0.3379497
80	79	0.3448065
81	80	0.3452172
82	81	0.3588692
83	82	0.3582511
84	83	0.3619837
85	84	0.3702216
86	85	0.3813289
87	86	0.3863549
88	87	0.3969943
89	88	0.4127410
90	89	0.4194072
91	90	0.4453816
92	91	0.4803600
93	92	0.5382100
94	93	0.5908958
95	94	0.6551174
96	95	0.7410796
97	96	0.8185894

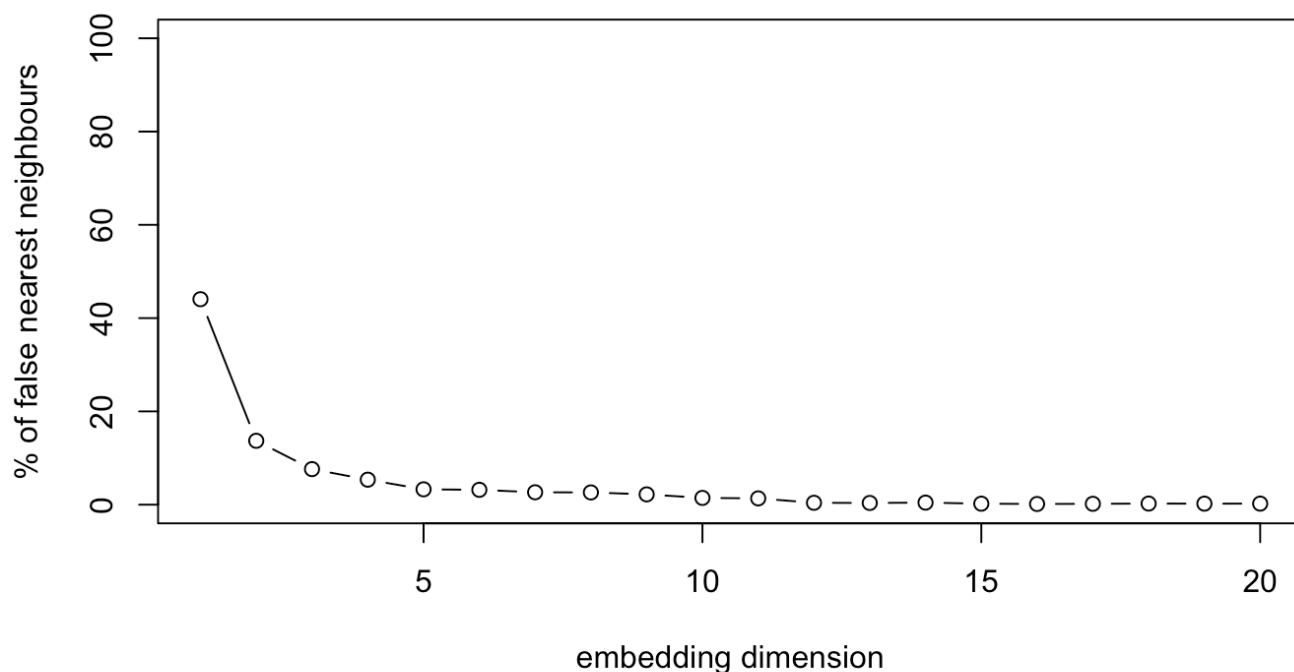
To estimate the delay parameter τ for RQA, Average Mutual Information (AMI) was computed for the scaled site 164 time series across lags 0 to 96. A maximum lag of 96 was chosen because each lag unit corresponds to one 15-minute interval, meaning 96 lags represents exactly one full day ($96 \times 15 = 1440$ minutes). This ensures the AMI search spans a complete daily cycle.

The AMI curve dropped sharply after lag 0 and did not show an early local minimum. This pattern is likely caused by the difference between very low night-time traffic volumes and high daytime peaks, which makes smaller within day patterns less visible in the mutual information plot. The first local minimum occurred at lag 29 (AMI = 0.378), corresponding to approximately 7.25 hr. This time interval matches the approximate separation between the morning and evening traffic peaks, which are the two main recurring traffic states at site 164.

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```
# Embedding (m)
fnn <- false.nearest(x_scaled, m = 20, d = 29, t = 1, eps = sd(x_scaled))
plot(fnn)
```

False nearest neighbours



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```
print(fnn)
```

	m1	m2	m3	m4	m5	m
6						
fraction	4.405679e-01	1.369773e-01	7.622807e-02	5.361154e-02	3.292085e-02	3.184526e-0
2						
total	3.774932e+06	1.656128e+06	1.003646e+06	6.464280e+05	4.373520e+05	3.189800e+0
5						
	m7	m8	m9	m10	m11	m1
2						
fraction	2.658207e-02	2.626588e-02	2.208446e-02	1.451956e-02	1.351351e-02	4.042862e-0
3						
total	2.225560e+05	1.602840e+05	1.187260e+05	8.650400e+04	6.704400e+04	5.095400e+0
4						
	m13	m14	m15	m16	m17	m1
8						
fraction	3.664273e-03	4.429951e-03	2.127827e-03	1.472176e-03	1.853454e-03	2.367051e-0
3						
total	4.039000e+04	3.250600e+04	2.537800e+04	2.037800e+04	1.618600e+04	1.267400e+0
4						
	m19	m20				
fraction	2.195609e-03	2.266432e-03				
total	1.002000e+04	7.942000e+03				

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```
embed <- 5
```

The embedding dimension, M , was estimated using the False Nearest Neighbors (FNN) method. This method identifies how many delayed versions of the time series are needed to unfold the underlying system dynamics. The FNN fraction was calculated for embedding dimensions from 1 to 20, using the previously selected delay of $\tau = 29$. The results showed a steep decrease from $M = 1$ (FNN = 44.0%) to $M = 2$ (FNN = 13.7%), followed by a slower decline at higher dimensions. After $M = 5$, where the FNN fraction reached 3.3%, further increases in embedding dimension produced only small improvements. Therefore, $M = 5$ was selected as a embedding dimension because it captures the main dynamics of the traffic system without adding unnecessary complexity.

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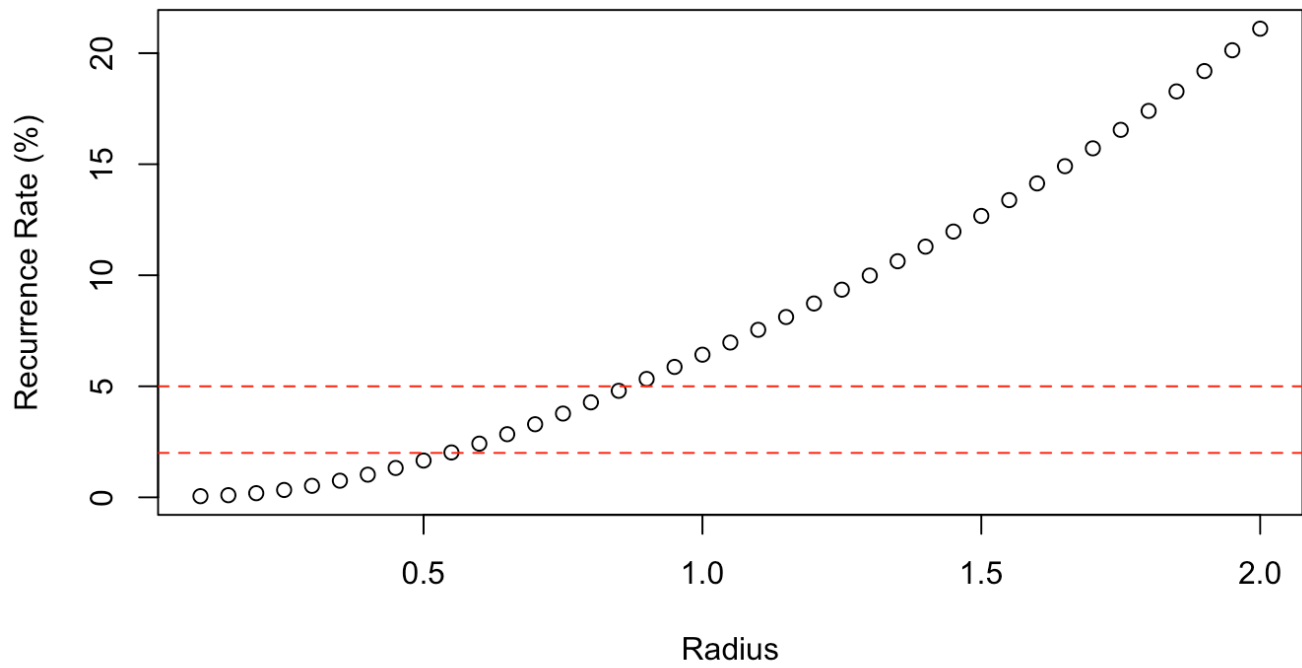
```
# Radius (r)

radius_candidates <- seq(0.1, 2.0, by = 0.05)
rr_values <- numeric(length(radius_candidates))

for (i in seq_along(radius_candidates)) {
  result <- crqa(
    ts1      = x_scaled,
    ts2      = x_scaled,
    delay     = 29,
    embed     = 5,
    radius    = radius_candidates[i]
  )
  rr_values[i] <- result$RR
}

plot(radius_candidates, rr_values, type = "b", xlab = "Radius", ylab = "Recurrence Rate (%)", main = "Recurrence Rate by Radius")
abline(h = c(2, 5), col = "red", lty = 2)
```

Recurrence Rate by Radius

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```
print(data.frame(radius = radius_candidates, RR = rr_values))
```

	radius	RR
1	0.10	0.05231642
2	0.15	0.09710287
3	0.20	0.18681626
4	0.25	0.33162766
5	0.30	0.51872490
6	0.35	0.74903515
7	0.40	1.02109740
8	0.45	1.32125654
9	0.50	1.65631202
10	0.55	2.02109403
11	0.60	2.41787839
12	0.65	2.84318111
13	0.70	3.29110183
14	0.75	3.77268262
15	0.80	4.27761194
16	0.85	4.79619635
17	0.90	5.33169510
18	0.95	5.87230747
19	1.00	6.42320332
20	1.05	6.97081182
21	1.10	7.54280843
22	1.15	8.11918816
23	1.20	8.72844125
24	1.25	9.35061892
25	1.30	9.98785652
26	1.35	10.63060112
27	1.40	11.28820897
28	1.45	11.96731095
29	1.50	12.67029529
30	1.55	13.38291686
31	1.60	14.13535174
32	1.65	14.91012365
33	1.70	15.71456588
34	1.75	16.54718930
35	1.80	17.39973342
36	1.85	18.27739616
37	1.90	19.19360784
38	1.95	20.13086410
39	2.00	21.10161186

The radius parameter was selected by inspecting recurrence rates across a range of candidate radius values from 0.10 to 2.00, computed in steps of 0.05. For each candidate radius, a recurrence quantification analysis was run using the previously determined parameters of $\tau = 29$ and $M = 5$, and the resulting recurrence rate was recorded.

Following the lecture guideline that continuous behavioral signals should ideally have a recurrence rate between 2% and 5%, with more deterministic signals placed closer to the lower end of this range, a radius of 0.55 was chosen. This radius produced a recurrence rate of 2.02%, indicating that the traffic volume time series at site 164 is highly structured and regular. This is consistent with its clear daily pattern, especially the repeated morning and evening traffic peaks.

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```
rqa_result <- crqa(  
  ts1      = x_scaled,  
  ts2      = x_scaled,  
  delay    = 29,  
  embed    = 5,  
  rescale  = 0,  
  radius   = 0.55,  
  normalize = 0,  
  mindiagline = 2,  
  minvertline = 2,  
  tw       = 29,  
  whiteline = FALSE,  
  recpt    = FALSE,  
  side     = "both"  
)  
  
cat("RR: ", rqa_result$RR, "\n")
```

RR: 1.790756

[Hide](#)

```
cat("DET: ", rqa_result$DET, "\n")
```

DET: 85.89001

[Hide](#)

```
cat("L: ", rqa_result$L, "\n")
```

L: 5.54574

[Hide](#)

```
cat("LAM: ", rqa_result$LAM, "\n")
```

LAM: 92.21856

[Hide](#)

```
cat("TT: ", rqa_result$TT, "\n")
```

TT: 4.147736

[Hide](#)

```
cat("ENTR:", rqa_result$ENTR, "\n")
```

ENTR: 2.091256

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```

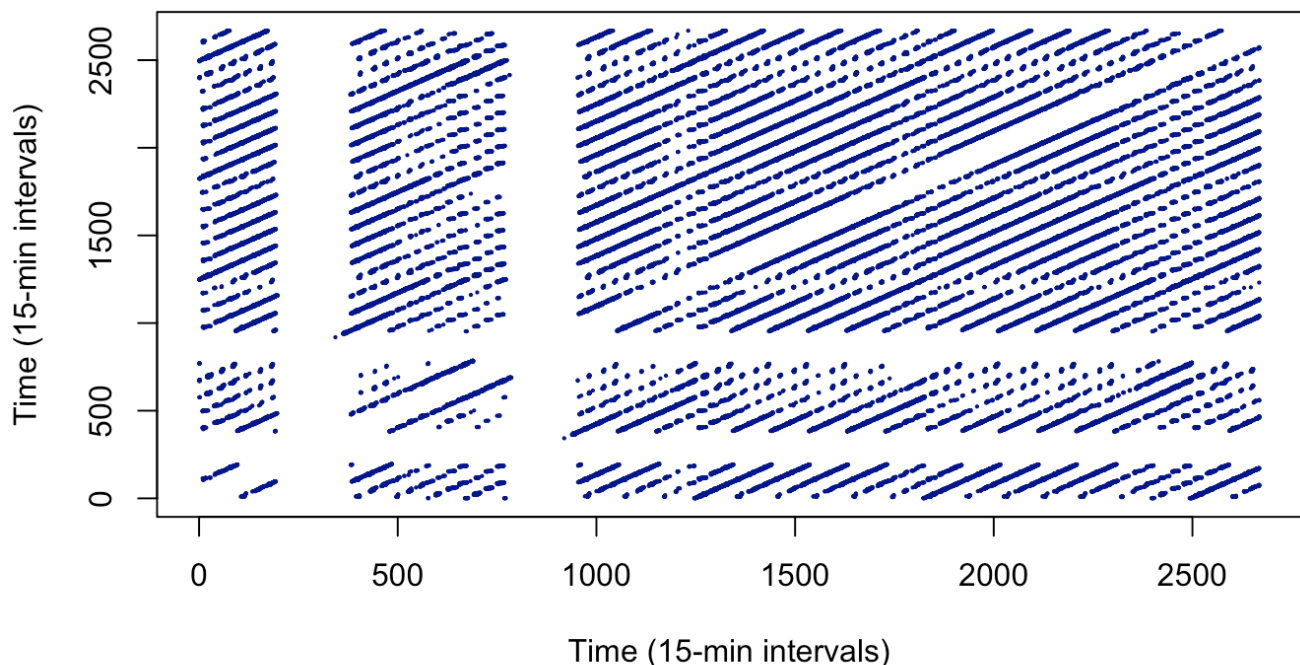
RP <- matrix(as.numeric(rqa_result$RP), nrow = nrow(rqa_result$RP))

plot(NULL, xlim = c(1, ncol(RP)), ylim = c(1, nrow(RP)),
      xlab = "Time (15-min intervals)",
      ylab = "Time (15-min intervals)",
      main = "Recurrence Plot - Site 164")

for (l in 1:ncol(RP)) {
  ind <- which(RP[, l] == 1)
  points(rep(l, length(ind)), ind,
        cex = 0.3,
        col = "blue4",
        pch = 16)
}

```

Recurrence Plot - Site 164



Prior to conducting the Recurrence Quantification Analysis (RQA), the time series of site 164 was standardized using z-score scaling. Consequently, the normalization and rescaling parameters within the `crqa()` function were both set to 0 (`normalize = 0`, `rescale = 0`), as the data had already been placed on a comparable scale.

For the final RQA, the previously obtained parameters were used: $\tau = 29$, $M = 5$, and `radius = 0.55`. The minimum diagonal line length and minimum vertical line length were both set to 2, as this is the minimum threshold required for a recurrent structure to be considered meaningful. The Theiler window was set to 29, matching the selected delay, to exclude points that were too close together in time and prevent trivial autocorrelation from being interpreted as meaningful recurrence. The analysis was conducted with `side = "both"`, meaning that recurrence structures on both sides of the main diagonal were included.

The RQA results indicated a recurrence rate of 1.79%, determinism of 85.89%, a mean diagonal line length of 5.55, laminarity of 92.22%, a trapping time of 4.15, and entropy of 2.09. The recurrence rate was slightly lower than the 2.02% obtained during radius selection, which is expected because the Theiler window (`tw = 29`) was applied in the final analysis to exclude recurrent points located within a short temporal interval of one another near the line of identity, thereby removing trivial autocorrelated recurrences from the calculation.

The high determinism value (DET = 85.89%) indicates that most recurrent points formed diagonal line structures, suggesting that traffic states often reoccurred in structured temporal sequences. The mean diagonal line length ($L = 5.55$) suggests that these recurring sequences persisted for an average of approximately 83 minutes (5.55×15 minutes) before diverging. Laminarity was notably high (LAM = 92.22%), even higher than the determinism value, indicating that a large proportion of recurrent points formed vertical structures. This suggests that the system often remained in stable or slowly changing traffic states for consecutive observations. This is further supported by the trapping time (TT = 4.15), corresponding to approximately 62 minutes of sustained state stability on average. The entropy value (ENTR = 2.09) indicates some variation in the lengths of recurring diagonal structures, suggesting that although traffic patterns were highly structured, they were not completely uniform. Taken together, these findings suggest that traffic at site 164 showed strong temporal structure and stability, consistent with repeated daily traffic rhythms rather than random fluctuations.

mdRQA - Multivariate Traffic System Analysis

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```
# Extract each site
site100 <- traffic_data[traffic_data$site_no == 100, ]
site161 <- traffic_data[traffic_data$site_no == 161, ]
site164 <- traffic_data[traffic_data$site_no == 164, ]

# Keep only timestamp and volume, and rename volume columns
site100 <- site100[, c("timestamp", "site_total_volume")]
site161 <- site161[, c("timestamp", "site_total_volume")]
site164 <- site164[, c("timestamp", "site_total_volume")]

colnames(site100)[2] <- "x100"
colnames(site161)[2] <- "x161"
colnames(site164)[2] <- "x164"

# Merge by shared timestamps only
traffic_joined <- merge(site100, site161, by = "timestamp")
traffic_joined <- merge(traffic_joined, site164, by = "timestamp")

# Order by timestamp
traffic_joined <- traffic_joined[order(traffic_joined$timestamp), ]

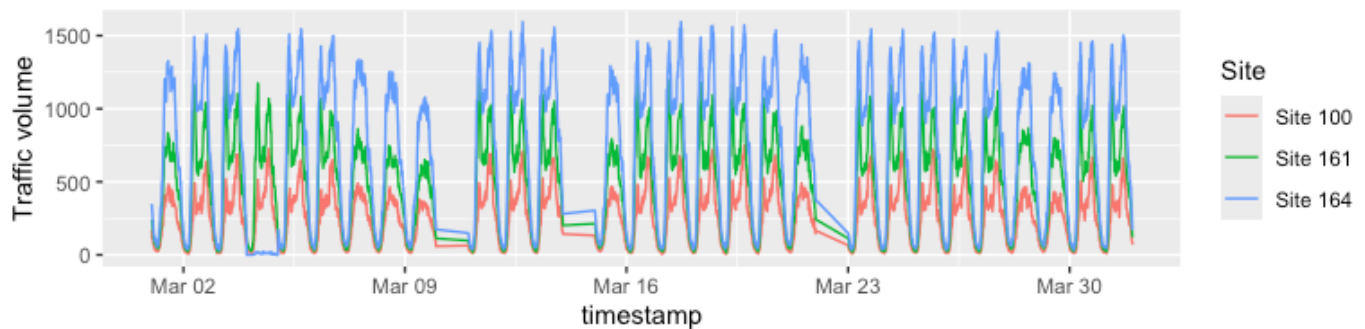
# Scale after alignment
traffic_matrix <- cbind(
  x100 = scale(traffic_joined$x100)[, 1],
  x161 = scale(traffic_joined$x161)[, 1],
  x164 = scale(traffic_joined$x164)[, 1]
)

# Check final dimensions
dim(traffic_matrix)
```

```
[1] 2688    3
```

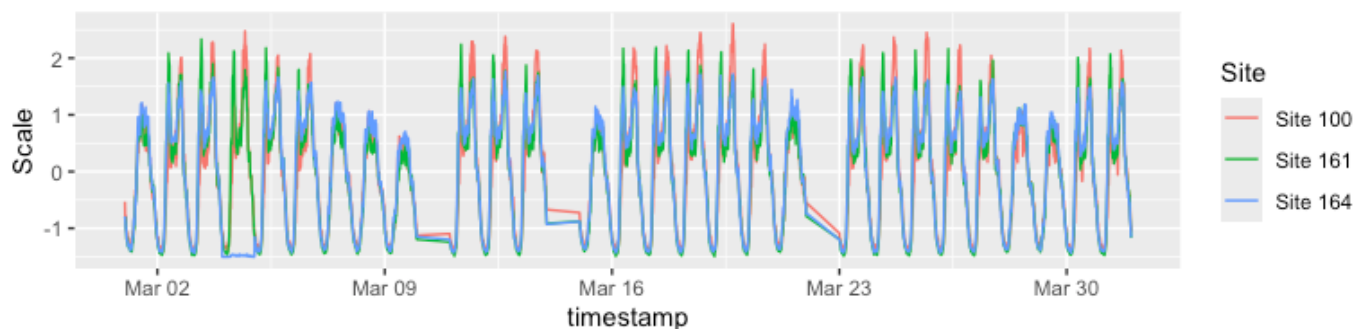
[Hide](#)

```
ggplot(traffic_joined, aes(x = timestamp)) +
  geom_line(aes(y = x100, color = "Site 100")) +
  geom_line(aes(y = x161, color = "Site 161")) +
  geom_line(aes(y = x164, color = "Site 164")) +
  labs(
    y = "Traffic volume",
    color = "Site"
  )
)
```



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```
ggplot(traffic_joined, aes(x = timestamp)) +
  geom_line(aes(y = scale(x100)[,1], color = "Site 100")) +
  geom_line(aes(y = scale(x161)[,1], color = "Site 161")) +
  geom_line(aes(y = scale(x164)[,1], color = "Site 164")) +
  labs(y = "Scale", color = "Site")
)
```



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```
# Convert to dataframe like they did
traffic_df <- as.data.frame(traffic_matrix)

nbins <- 10
# one full day
maxlag <- 96
criterion <- "firstBelow"
threshold <- exp(-1)

md_delay <- mdDelay(traffic_df, nbins, maxlag, criterion, threshold)
print(md_delay)
```

```
[1] 21
```

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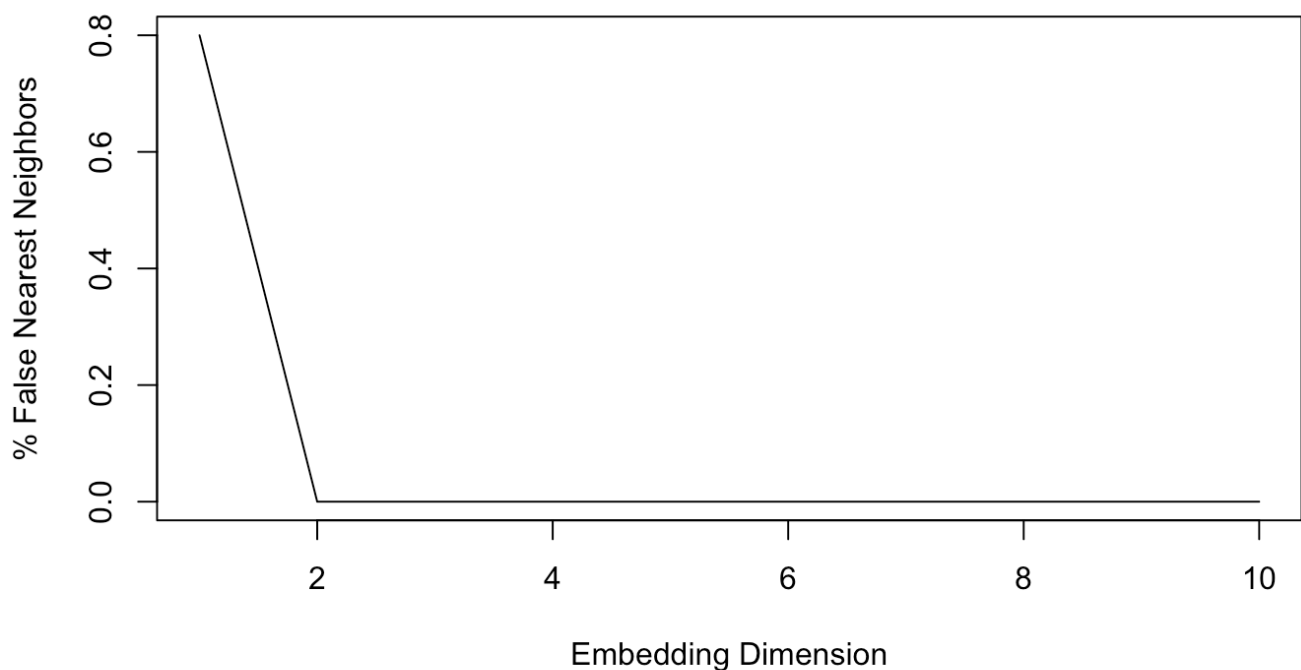
```

tau <- md_delay
maxEmb <- 10
numSamples <- 500
Rtol <- 10
Atol <- 2

md_fnn <- mdFnn(traffic_df, tau, maxEmb, numSamples, Rtol, Atol)
plot(md_fnn$embTimes[1:length(md_fnn$fnnPerc)], md_fnn$fnnPerc,
     type = 'l',
     xlab = "Embedding Dimension",
     ylab = "% False Nearest Neighbors",
     main = "mdFNN - All Three Sites")

```

mdFNN - All Three Sites


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```
print(md_fnn)
```

```

$fnnPerc
[1] 0.8 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0

$embTimes
[1] 1 2 3 4 5 6 7 8 9 10 11

```

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```
md_embed <- 2
```

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```

par <- list(method = "mdcrqa", metric = "euclidean", maxlag = 96,
            radiusspan = 100, radiussample = 40, normalize = 0,
            rescale = 0, mindiagline = 2, minvertline = 2, tw = 0,
            whiteline = FALSE, recpt = FALSE, side = "both",
            datatype = "continuous", fnnpercent = NA,
            typeami = NA, nbins = 50, criterion = "firstBelow",
            threshold = 1, maxEmb = 20, numSamples = 500,
            Rtol = 10, Atol = 2)

results <- optimizeParam(traffic_df, traffic_df, par, min.rec = 2, max.rec = 5)
print(unlist(results))

```

radius	emddim	delay
0.5490443	2.0000000	16.0000000

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```

delay <- md_delay
embed <- md_embed
rescale <- 0
radius <- results$radius
normalize <- 0
mindiagline <- 2
minvertline <- 2
tw <- 0
whiteline <- FALSE
recpt <- FALSE
side <- "both"
method <- "mdcrqa"
metric <- "euclidean"
datatype <- "continuous"

mdrqa_result <- crqa(traffic_df, traffic_df, delay, embed, rescale, radius,
                    normalize, mindiagline, minvertline, tw,
                    whiteline, recpt, side, method, metric, datatype)

cat("RR: ", mdrqa_result$RR, "\n")

```

RR: 3.524855

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```
cat("DET: ", mdrqa_result$DET, "\n")
```

DET: 83.23701

Hide

```
cat("L: ", mdrqa_result$L, "\n")
```

L: 5.024945

Hide

```
cat("LAM: ", mdrqa_result$LAM, "\n")
```

```
LAM: 90.43311
```

Hide

```
cat("TT: ", mdrqa_result$TT, "\n")
```

```
TT: 4.421557
```

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```
cat("ENTR:", mdrqa_result$ENTR, "\n")
```

```
ENTR: 2.091264
```

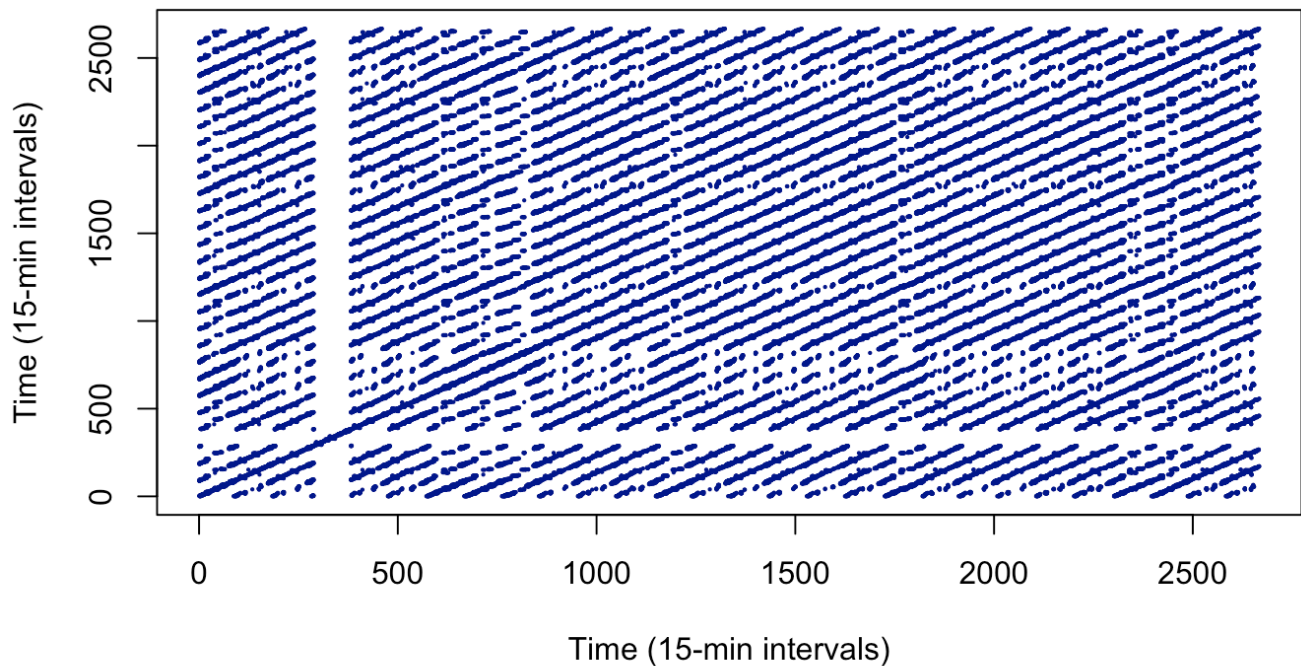
Hide

```
RP <- matrix(as.numeric(mdrqa_result$RP), nrow = nrow(mdrqa_result$RP))

plot(NULL, xlim = c(1, ncol(RP)), ylim = c(1, nrow(RP)),
      xlab = "Time (15-min intervals)",
      ylab = "Time (15-min intervals)",
      main = "Recurrence Plot - All Sites (mdRQA)")

for (l in 1:ncol(RP)) {
  ind <- which(RP[, l] == 1)
  points(rep(l, length(ind)), ind,
         cex = 0.3, col = "blue4", pch = 16)
}
```

Recurrence Plot - All Sites (mdRQA)

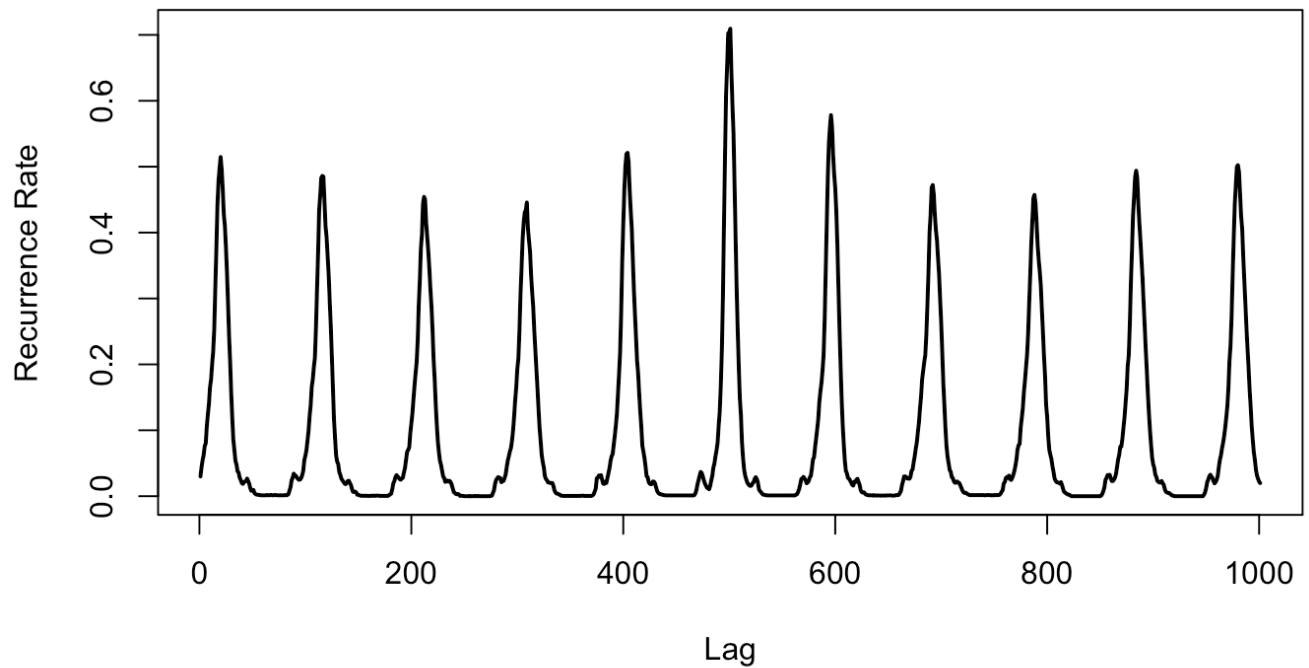

[Hide](#)

```
res <- drpfrofromts(
  traffic_df[, 1],
  traffic_df[, 3],
  windowsize = 500,
  radius = results$radius,
  delay = md_delay,
  embed = md_embed,
  rescale = 0,
  normalize = 0,
  mindiagline = 2,
  minvertline = 2,
  tw = 0,
  whiteline = FALSE,
  recpt = FALSE,
  side = "both",
  method = "crqa",
  metric = "euclidean",
  datatype = "continuous"
)

profile <- res$profile

plot(seq(1, length(profile), 1), profile,
     type = "l", lwd = 2,
     xlab = "Lag",
     ylab = "Recurrence Rate",
     main = "Diagonal Cross-Recurrence Profile - Site 100 vs Site 164")
```

Diagonal Cross-Recurrence Profile - Site 100 vs Site 164



TODO: Improve DCRP plot TODO: REPORT THE RESULTS OF mdRQA TODO: Justify the selected parameters (Similar to RQA)